HOPPER INFORMATION SERIES

08 / FIDUCIAL MARKERS

AprilTags with Hopper

Visual IDs for detection, centering and 2D heading alignment.

tag36h11

tag36h11 ID 0

ID + orientation

AprilTags were built to give robots reliable visual landmarks

2011
AprilTag introduced

Edwin Olson, University of Michigan APRIL Lab; ICRA paper.

2016
AprilTag 2

A redesigned detector improved speed, detection rate, and false-positive behavior.

AprilTag 3
Faster + flexible

Small-tag detection, pose estimation, and more layouts/families.

TODAY
Robotics standard

Open implementations in research, ROS, calibration, navigation, and labs.

The original goal: print an artificial feature that a camera can identify and localize—even when natural visual features are unreliable.

A printed tag can become a shared coordinate reference

LOCALIZATION

• Estimate a robot or camera pose relative to a known landmark.

NAVIGATION + DOCKING

• Center on a landing pad, charging station, doorway, or work cell.

CALIBRATION

• Relate cameras, robot frames, and measurement rigs with known geometry.

MANIPULATION

• Identify bins, tools, parts, or target objects for a robot arm.

SLAM + GROUND TRUTH

• Supply repeatable landmarks while testing perception and state estimation.

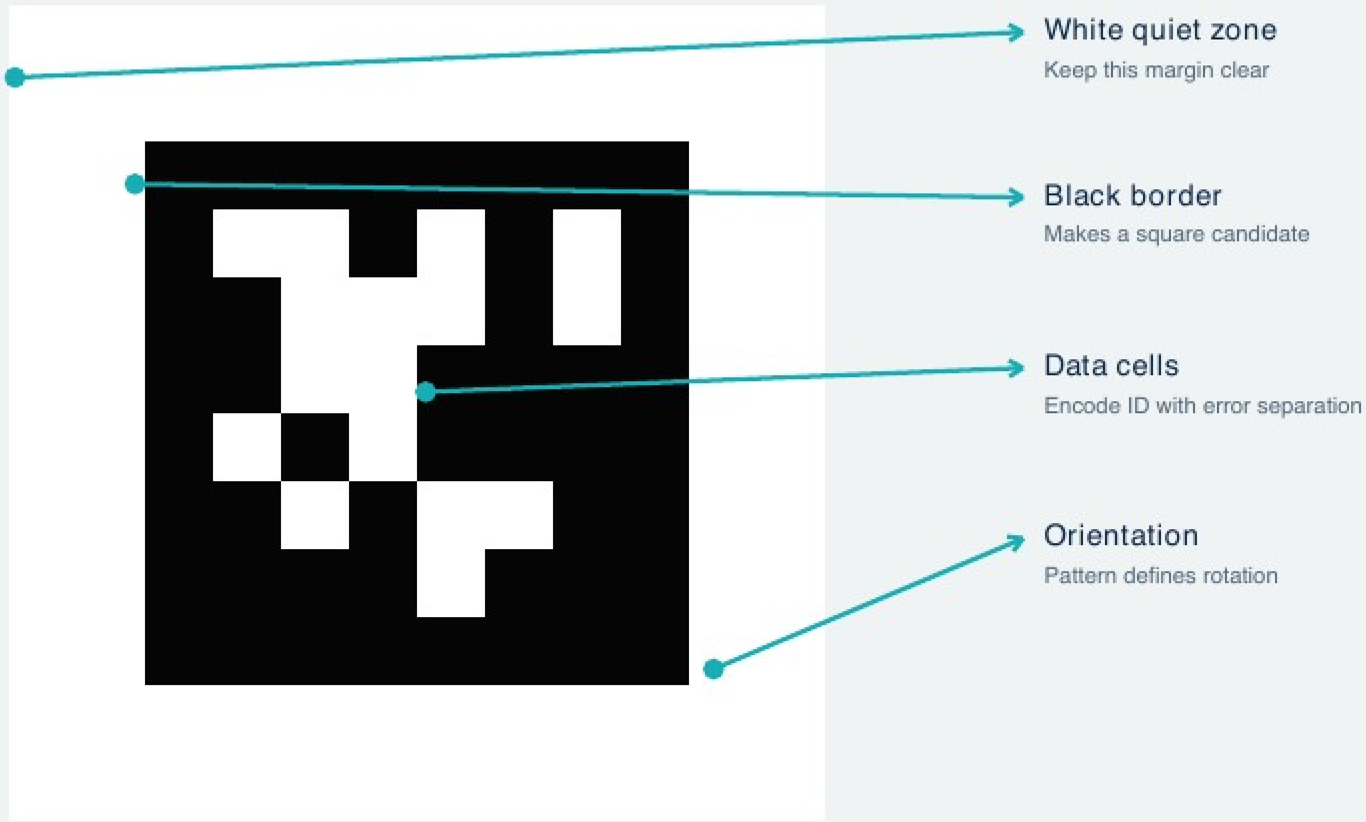
AR + HUMAN INTERFACES

• Anchor digital information to a physical place or object.

A tag is useful because its identity and geometry are designed—not discovered by chance.

TAG ANATOMY

A visual ID designed for robots



tag36h11 ID 0

HOPPER FAMILY

- tag36h11
- IDs 0 through 586
- Black and white only
- Up to 3 payload-bit errors accepted

WHY FIDUCIALS

- Known geometry.
- Unique machine-readable ID.
- Orientation from one image.
- No battery or radio on the tag.

Family chooses the codebook; ID chooses one tag inside it

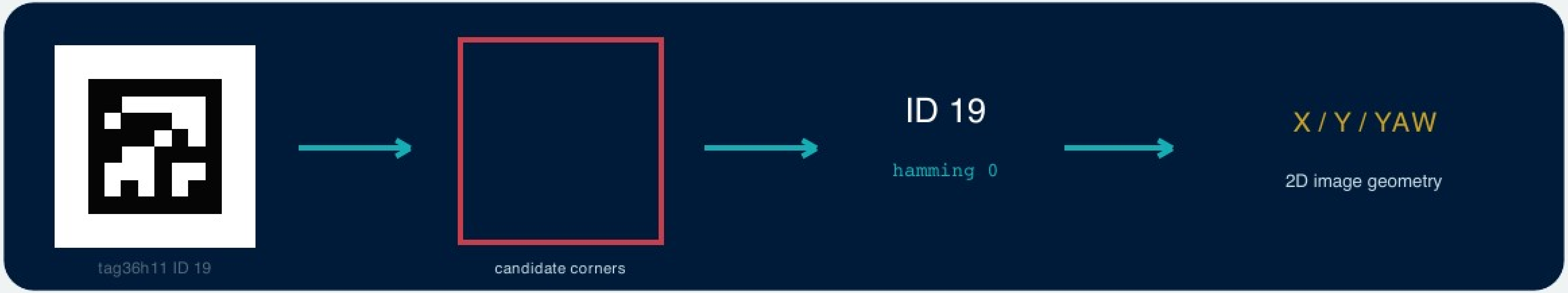
FAMILY	CODE DESIGN	WHEN IT FITS
tag16h5	Small code, fewer IDs	Legacy / compact demonstrations
tag25h9	More bits and separation	Legacy balance of size and robustness
tag36h11	36 data bits · minimum Hamming distance 11	Hopper Studio: IDs 0–586
tagStandard41h12	Modern standard layout	Upstream default for most new work
tagCircle49h12	Circular, space-efficient layout	Small circular objects

Never mix families in one detector configuration: the same ID number in another family is a different code.

LOCAL TAG36H11 PIPELINE

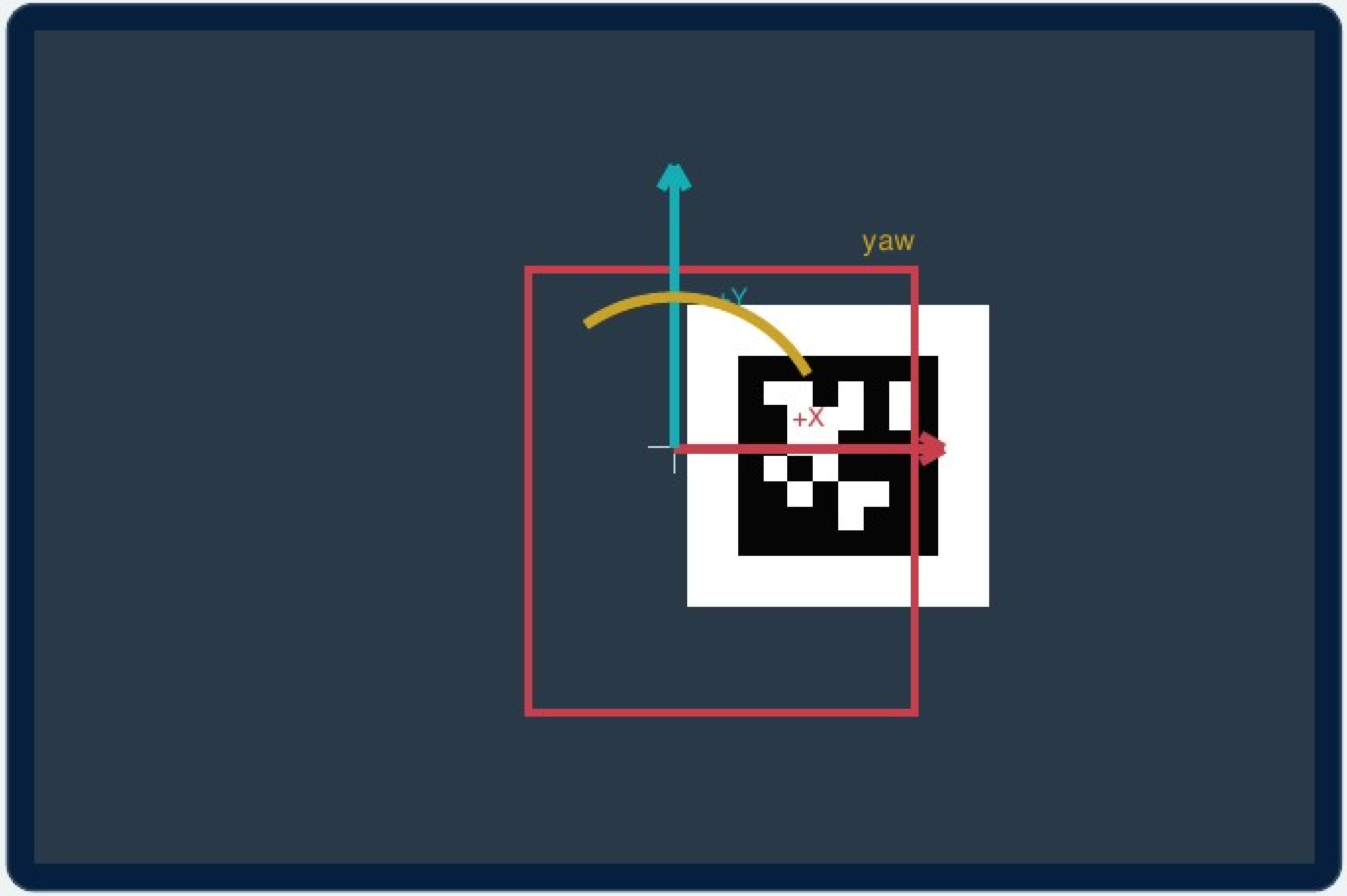
How Hopper Studio detects a tag

1. **Grayscale** — Convert RGB camera pixels to luminance.
2. **Threshold** — Estimate a black/white split.
3. **Squares** — Find connected dark square candidates.
4. **Sample** — Map each candidate into an 8x8 grid.
5. **Match** — Compare rotations with the codebook.
6. **Pose** — Return ID, corners, center and 2D yaw.



WHAT THE RESULT MEANS

Coordinates and pose in Hopper Studio



APRILTAG RESULT

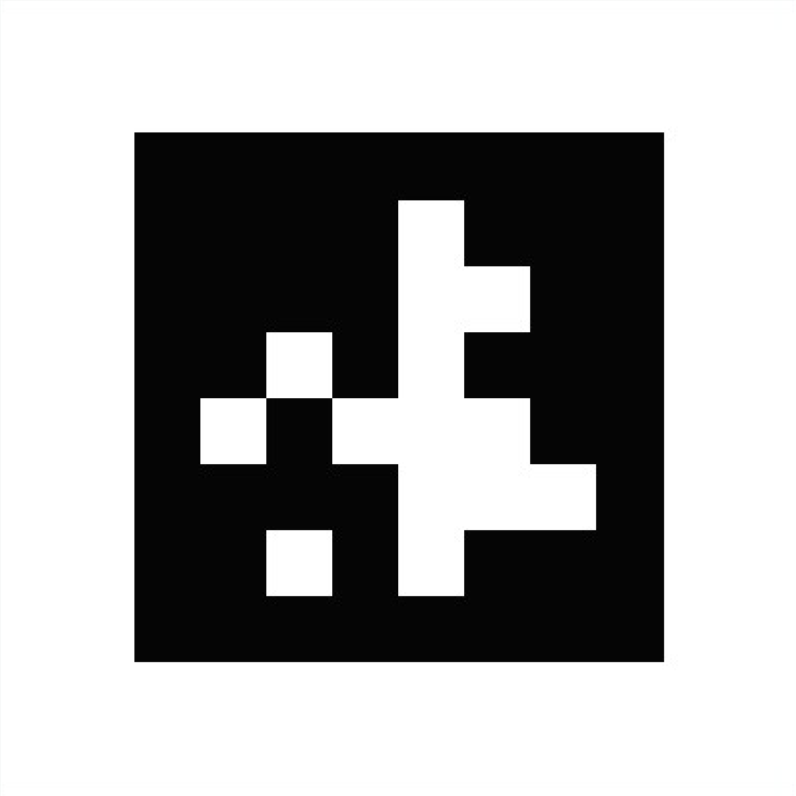
- id + family + hamming
- four corners + center in pixels
- bounding box
- centerX / centerY: -100..100
- yaw: normalized around +/-180 deg

DO NOT OVERCLAIM

- Yaw is 2D image-plane orientation.
- It is not full 3D pose or aircraft heading.
- No distance or altitude is returned.
- centerOnAprilTag never lands.

PHYSICAL TAG SETUP

Print cleanly and place deliberately



tag36h11 ID 7

PRINT IN HOPPER STUDIO

- Vision Testing -> AprilTag Detection.
- Choose an ID from 0-586.
- Generate PDF for a full-page US Letter vector tag.
- Print at 100% scale with a clean white margin.

PLACE FOR THE CAMERA

- Keep the sheet flat and well lit.
- Avoid glare, folds, shadows and clipped margins.
- Start large and close; test before increasing height.
- Keep paper and people clear of propellers.

From a printed ID to a mission decision

01 Generate
Choose ID 0–586 and export the printable PDF.

02 Place
Keep the full border and quiet zone visible.

03 Scan
Test once or use scan for AprilTags.

04 Interpret
Read ID, center x/y, and 2D yaw.

05 Act
Center, align, branch, time out, or land safely.

Block patterns

- scan for AprilTags
- camera sees AprilTag with ID any / 0–586
- center on AprilTag with power, center slack, angle slack, and lost-tag limit

Hopper Studio provides a relative visual landmark—not global position, altitude, or collision avoidance.

CHECKLIST AND SOURCES

Reliable tags need reliable experiments

BEFORE FLIGHT

- Confirm the family is tag36h11.
- Test the intended ID, size, angle and lighting.
- Keep the full border and quiet zone in frame.
- Use a lost-tag limit and an overall timeout.
- Plan a safe fallback land.

SOURCES AND FURTHER READING

University of Michigan APRIL Lab - AprilTag
https://april.eecs.umich.edu/software/apriltag

AprilTag 2 paper
https://april.eecs.umich.edu/media/pdfs/wang2016iros.pdf

Hopper Studio detector and printable tag generator
lib/apriltags.ts, lib/vision.ts and README.md

DETECT -> CENTER -> ALIGN -> DECIDE

The mission still owns height, obstacle clearance and landing.